

Executive Data Analysis Report

Project Identifier: AI-DATA-ANALYST-20260726-141704

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1. Executive Summary

The model achieves an overall accuracy of 74.9% across all orders, meaning it correctly distinguishes instances of late delivery from on-time deliveries just over three-quarters of the time. To rank and separate orders, the model achieves an ROC-AUC of 0.781, indicating moderate ability to tell apart orders destined for late delivery from those arriving on time. Using a standard decision threshold of 0.5, the model correctly flags 1,043 orders as cases of late delivery. However, this same threshold misses 538 orders that were actually late delivery, predicting them as on-time. Additionally, 4,291 orders that were actually on-time delivery get incorrectly flagged as cases of late delivery, creating false alarms. The remaining 13,397 orders are correctly identified as on-time, while the 1,043 correctly flagged cases of late delivery show the model's correct calls at this threshold.

2. Data Diagnostics Profile

Input shape	96,476 rows × 22 columns
Output shape	96,476 rows × 22 columns
Duplicate rows removed	0
High-missing columns flagged	None
Low-variance columns dropped	None
Columns type-corrected	None
Numeric columns imputed	customer_zip_code_prefix, n_items, n_distinct_sellers, n_distinct_products, total_price, total_freight, avg_product_weight_g, avg_product_photos_qty, total_payment_value, max_installments, n_payment_methods, purchase_to_estimated_days, is_late_delivery
Categorical columns imputed	order_id, order_purchase_timestamp, order_estimated_delivery_date, customer_unique_id, customer_city, customer_state, primary_seller_state, product_category, primary_payment_type

3. Model Performance Leaderboard

Task type: binary_classification | **Best model:** HistGradientBoostingClassifier

Candidate Model	CV Score
HistGradientBoostingClassifier	0.5693
RandomForestClassifier	0.5632
LogisticRegression	0.4738

F1 (macro)	0.5745
Precision (macro)	0.5785
Recall (macro)	0.7086
ROC-AUC	0.7807

	Predicted Negative (0)	Predicted Positive (1)
Actual Negative (0)	13,397	4,291
Actual Positive (1)	538	1,043

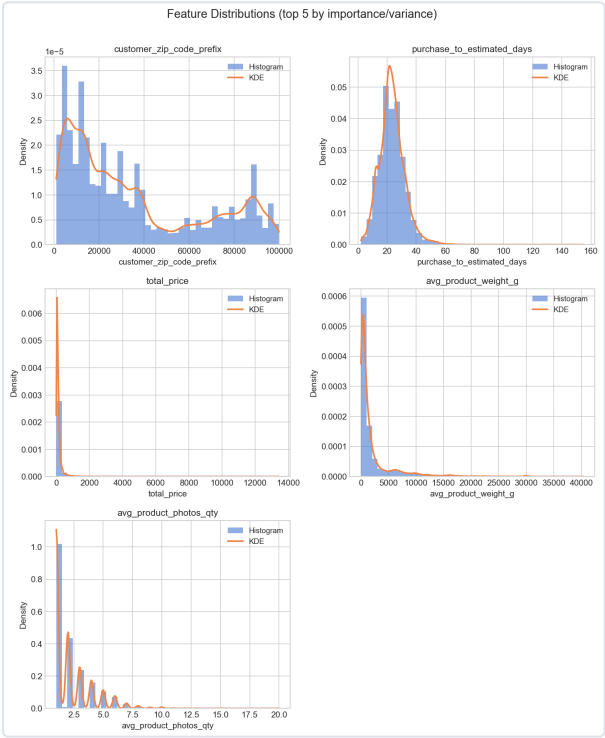
Threshold	Precision	Recall	F1 (macro)
0.3	0.122	0.908	0.399
0.4	0.154	0.796	0.503
0.5	0.196	0.660	0.574

Note on probabilities: this model uses `class_weight='balanced'` to handle class imbalance, which systematically shifts its raw predicted probabilities away from true likelihoods. Treat each decision threshold above (e.g. `threshold=0.3`) as a precision/recall cutoff chosen for a business tradeoff, not a literal percentage chance of the outcome. Measured calibration error (ECE) for this model: 0.306.

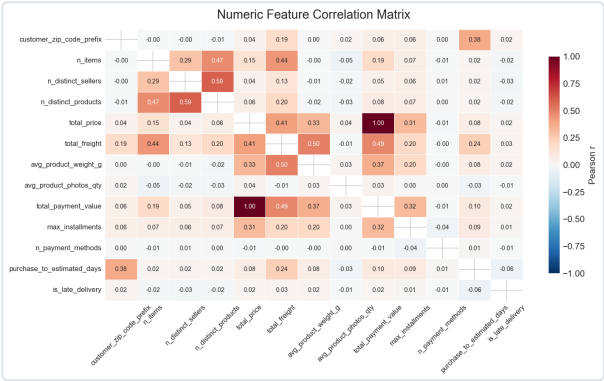
This model uses `class_weight='balanced'` to handle class imbalance, which systematically shifts `predict_proba` outputs away from true probabilities -- exactly what a calibration curve measures. Treat a decision threshold (e.g. `threshold=0.3` in the threshold sweep) as a precision/recall cutoff chosen for a business tradeoff, not a literal percentage chance of the outcome. Measured ECE = 0.3061 exceeded the 0.05 threshold, so a isotonic-calibrated comparison model was fit ('HistGradientBoostingClassifier' + CalibratedClassifierCV, `cv=3`). Its ECE = 0.0070 vs. the raw model's 0.3061 (lower is better). This comparison is diagnostic only -- the serialized production model is still the raw, uncalibrated estimator.

4. Automated Business Insights

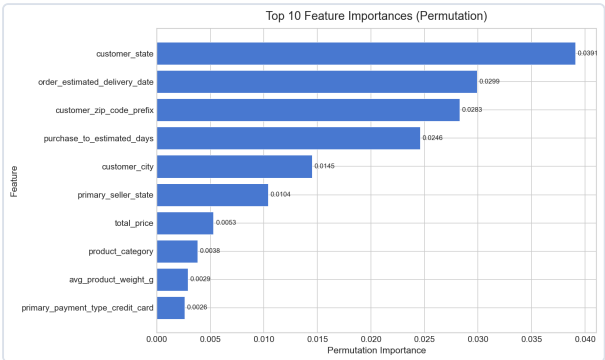
The state where the customer operates matters most for predicting cases of late delivery, as different states likely introduce varying logistics challenges that cause cases of late delivery for orders shipped to those regions. The expected arrival date assigned to an order is the second most important factor, meaning orders with certain estimated delivery dates face higher risks of cases of late delivery than others. Specific geographic zip prefixes within a customer state strongly predict instances of late delivery, because certain zip code ranges likely indicate remote or congested areas that lead to delayed orders. The number of days between the purchase and the estimated delivery date affects whether an order experiences cases of late delivery, as shorter promised windows leave less room for error and increase the chances of late delivery. The specific city where an order is destined or originates helps differentiate cases of late delivery, since urban versus rural routes create distinct delay risks that cause orders to arrive late. The state where the seller is located influences late delivery cases because mismatches between the seller state and customer state impact the logistics timeline and whether an order arrives on time.



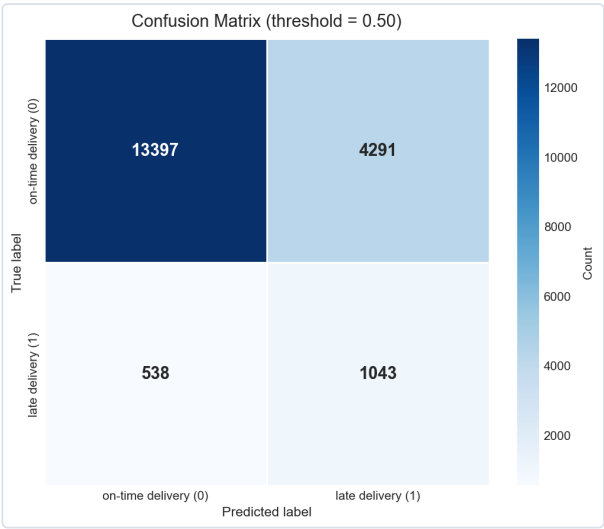
Distributions



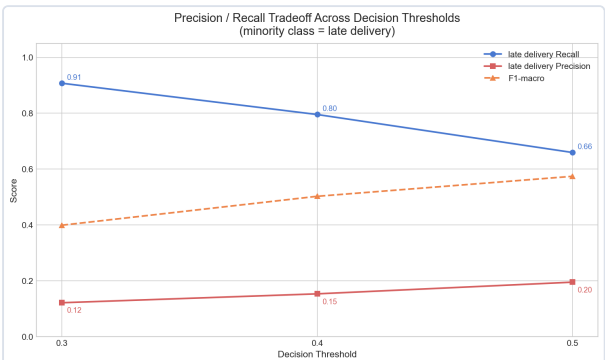
Correlation Heatmap



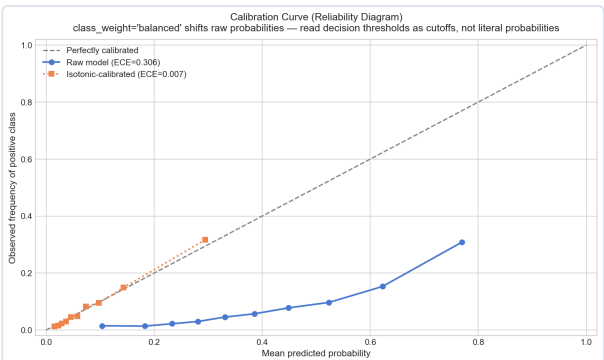
Feature Importance



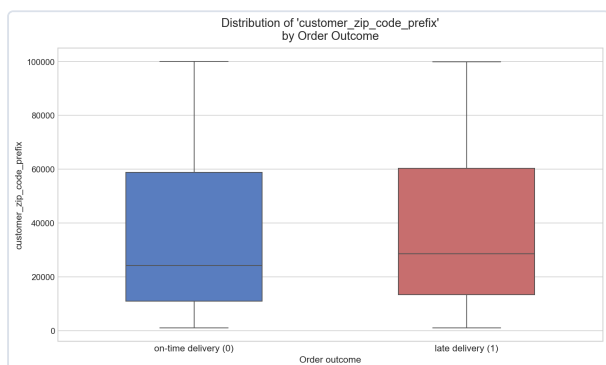
Confusion Matrix



Threshold Tradeoff



Calibration Curve



Top Feature Vs Target

5. Strategic Recommendations

To prioritize catching every possible instance of late delivery, set the threshold to 0.3. At this threshold, the model achieves a recall of 0.908, meaning it catches 90.8% of cases of late delivery before they occur, but precision is only 0.122, so the vast majority of orders flagged as late delivery will be false alarms requiring manual verification. To balance catching late delivery orders with operational efficiency, set the threshold to 0.4. This provides a recall of 0.796, catching 79.6% of late delivery instances, while precision rises to 0.154, meaning 15.4% of orders flagged as late delivery are genuine cases, reducing the manual review burden compared to the 0.3 threshold. If the business goal is to ensure high confidence in the orders flagged as late delivery, set the threshold to 0.5. At this point, precision reaches 0.196, meaning 19.6% of flagged orders are true cases of late delivery, but recall drops to 0.660, causing the model to miss 34.0% of actual late delivery orders.